PLS Midterm Project

Trixie Southwood

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Load in libraries

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.1     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(maps)
```

```
##
## Attaching package: 'maps'
##
## The following object is masked from 'package:purrr':
##
##     map
```

```r
library(mapproj)
library(lubridate)
library(viridis)
```

```
## Loading required package: viridisLite
##
## Attaching package: 'viridis'
##
## The following object is masked from 'package:maps':
##
##     unemp
```

Read in data

```r
ccc1 <- read.csv("ccc_compiled_20172020.csv")
ccc1 <- ccc1 %>%
  select(-starts_with("source"))


ccc2 <- read.csv("ccc_compiled_20212024.csv")
ccc2 <- ccc2 %>%
  select(-starts_with("source"))

ccc <- bind_rows(ccc1, ccc2)
```

Filter by issue

```r
ccc_sep_issues <- ccc %>%
  separate_longer_delim(issues, delim = ";")
```

Filter by vigil

```r
vigils <- ccc %>%
  filter(type == "vigil")

head(vigils)
```

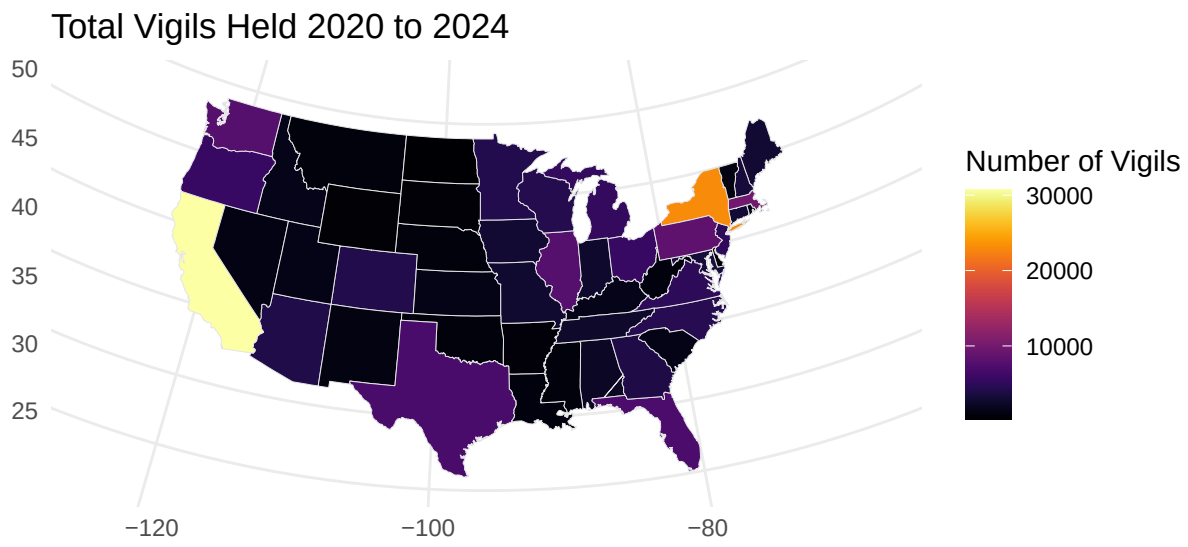
```
##         date       locality state                           location_detail
## 1 2017-01-01     Washington    DC                      Lafayette Square Park
## 2 2017-01-01        Mankato    MN                                       <NA>
## 3 2017-01-01 Little Compton    RI                                  Town Green
## 4 2017-01-01      Oak Ridge    TN              Y-12 National Security Complex
## 5 2017-01-02     Washington    DC                      Lafayette Square Park
## 6 2017-01-02         Delmar    NY Four Corners, Kenwood Ave and Delaware Ave
##   online  type macroevent                                actors
## 1      0 vigil       <NA>
## 2      0 vigil       <NA>                    Peace Vigil Mankato
## 3      0 vigil       <NA>                Sakonnet Peace Alliance
## 4      0 vigil       <NA> Oak Ridge Environmental Peace Alliance
## 5      0 vigil       <NA>
## 6      0 vigil       <NA>          Bethlehem Neighbors for Peace
##                                          claims valence
## 1         for banning nuclear weapons, for peace       0
## 2                                     for peace       0
## 3 for peace, for gun control, for climate action       1
## 4                 for abolishing nuclear weapons       0
## 5         for banning nuclear weapons, for peace       0
## 6                         for peace, against war       0
##                        issues size_text size_low size_high size_mean size_cat
## 1                    military      <NA>       NA        NA        NA        0
## 2                    military      <NA>       NA        NA        NA        0
## 3 environment; guns; military      <NA>       NA        NA        NA        0
## 4                    military      <NA>       NA        NA        NA        0
## 5                    military      <NA>       NA        NA        NA        0
## 6                    military      <NA>       NA        NA        NA        0
##   arrests arrests_any injuries_crowd injuries_crowd_any injuries_police
```

```
## 1    <NA>           0           <NA>                  0            <NA>
## 2    <NA>           0           <NA>                  0            <NA>
## 3    <NA>           0           <NA>                  0            <NA>
## 4    <NA>           0           <NA>                  0            <NA>
## 5    <NA>           0           <NA>                  0            <NA>
## 6    <NA>           0           <NA>                  0            <NA>
##   injuries_police_any property_damage property_damage_any chemical_agents
## 1                   0            <NA>                   0               0
## 2                   0            <NA>                   0               0
## 3                   0            <NA>                   0               0
## 4                   0            <NA>                   0               0
## 5                   0            <NA>                   0               0
## 6                   0            <NA>                   0               0
##                                                   notes      lat       lon
## 1 White House Peace Vigil continuous since June 3, 1981 38.90719 -77.03687
## 2                              every Sunday since 2001 44.16358 -93.99940
## 3                              every Sunday since 2003 41.51010 -71.17116
## 4                     every Sunday since the late 1990s 36.01036 -84.26964
## 5 White House Peace Vigil continuous since June 3, 1981 38.90719 -77.03687
## 6                              every Monday since 2003 42.62202 -73.83262
##   resolved_locality      resolved_county resolved_state fips_code title
## 1        Washington District of Columbia             DC     11001  <NA>
## 2           Mankato    Blue Earth County             MN     27013  <NA>
## 3    Little Compton       Newport County             RI     44005  <NA>
## 4         Oak Ridge      Anderson County             TN     47001  <NA>
## 5        Washington District of Columbia             DC     11001  <NA>
## 6            Delmar        Albany County             NY     36001  <NA>
##   organizations participants claims_summary claims_verbatim issue_tags_summary
## 1          <NA>         <NA>           <NA>            <NA>               <NA>
## 2          <NA>         <NA>           <NA>            <NA>               <NA>
## 3          <NA>         <NA>           <NA>            <NA>               <NA>
## 4          <NA>         <NA>           <NA>            <NA>               <NA>
## 5          <NA>         <NA>           <NA>            <NA>               <NA>
## 6          <NA>         <NA>           <NA>            <NA>               <NA>
##   issue_tags_verbatim issue_tags participant_measures police_measures
## 1                <NA>       <NA>                 <NA>            <NA>
## 2                <NA>       <NA>                 <NA>            <NA>
## 3                <NA>       <NA>                 <NA>            <NA>
## 4                <NA>       <NA>                 <NA>            <NA>
## 5                <NA>       <NA>                 <NA>            <NA>
## 6                <NA>       <NA>                 <NA>            <NA>
##   participant_deaths police_deaths
## 1               <NA>          <NA>
## 2               <NA>          <NA>
## 3               <NA>          <NA>
## 4               <NA>          <NA>
## 5               <NA>          <NA>
## 6               <NA>          <NA>
```

I expected guns to have the highest levels of vigils, however it is military. So lets plot monthly military vigils compared with monthly gun vigils

```r
state_abbr_to_name <- data.frame(
  state_abbr = c(state.abb, "DC"),
  state_name = c(tolower(state.name), "District of Columbia")
)

state_vigil_counts <- ccc %>%
  count(state)

state_vigil_counts <- state_vigil_counts %>%
  left_join(state_abbr_to_name, by = c("state" = "state_abbr"))

us_states <- map_data("state")

dc_map_data <- map_data("state") %>%
  filter(region == "district of columbia") %>%
  mutate(region = "district of columbia") %>%
  bind_rows(data.frame(
    region = "district of columbia",
    long = -77.0369,
    lat = 38.9072,
    group = 1,
    order = 1
  ))

us_states <- bind_rows(us_states, dc_map_data)

us_states_vigils <- us_states %>%
  left_join(state_vigil_counts, by = c("region" = "state_name"))

label_data <- us_states_vigils %>%
  filter(region %in% c("california", "new york", "district of columbia"))

ggplot(data = us_states_vigils, mapping = aes(x = long, y = lat, group = group, fill = n)) +
  geom_polygon(color = "gray90", size = 0.1) +
  coord_map(projection = "albers", lat0 = 39, lat1 = 45) +
  scale_fill_viridis(option = "inferno", name = "Number of Vigils", na.value = "gray") +
  labs(title = "Total Vigils Held 2020 to 2024", x = " ", y = " ") +
  theme_minimal()
```

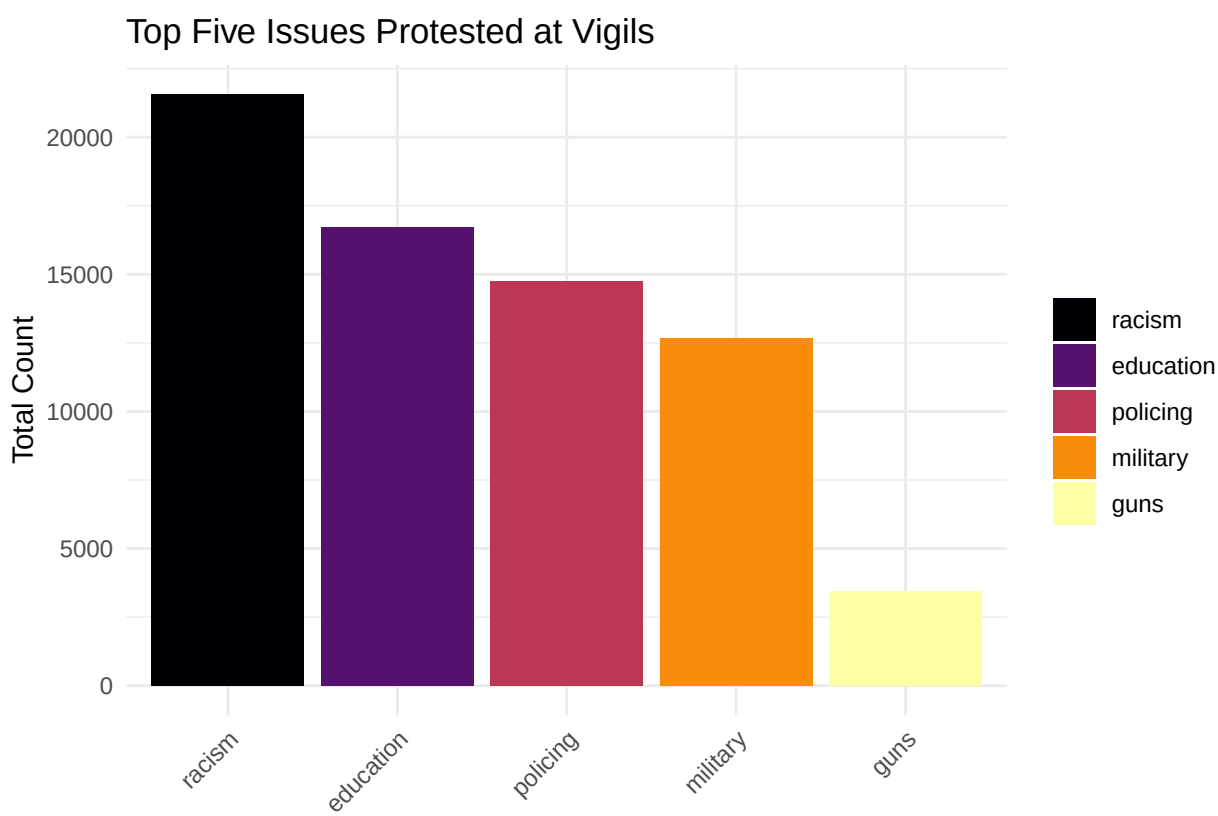
```
## Warning: Using ‘size‘ aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use ‘linewidth‘ instead.
## This warning is displayed once every 8 hours.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```




```
  ggsave("vigils_by_state_map.png", width = 10, height = 8, dpi = 300)
```

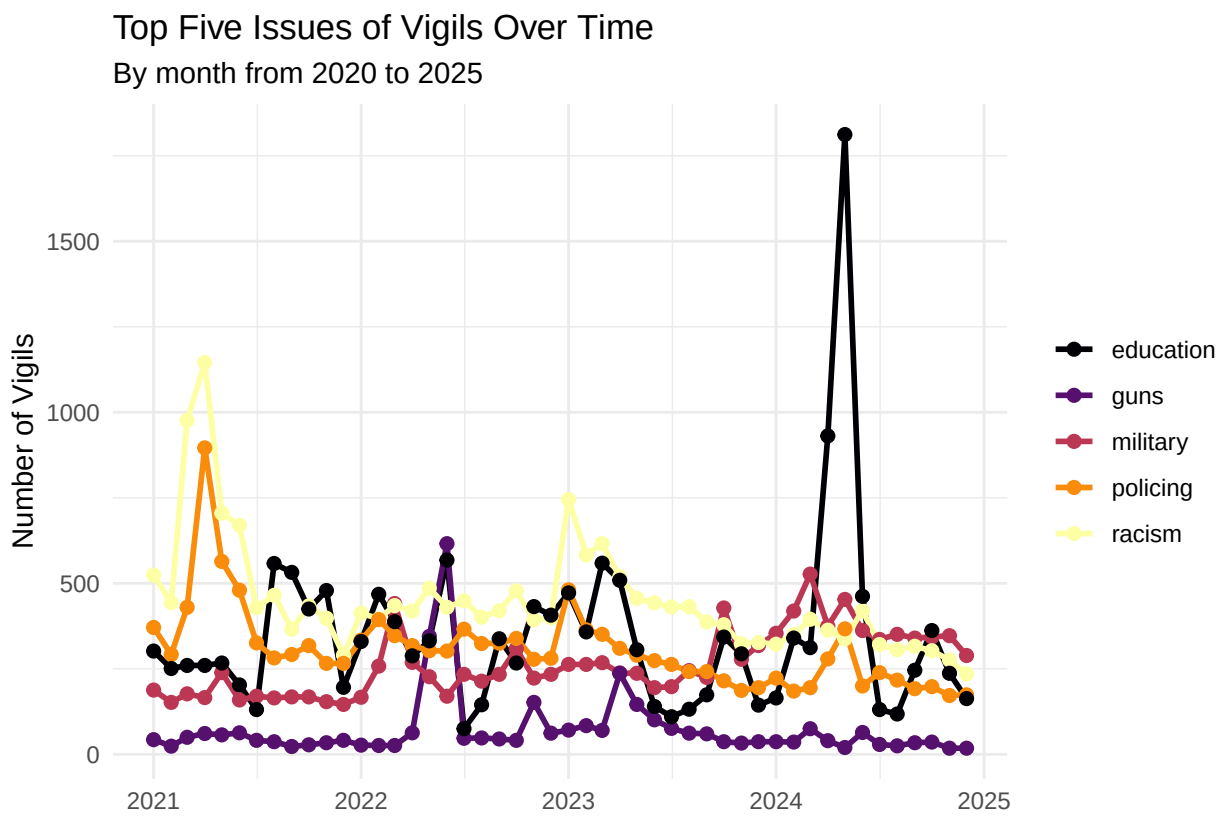
```
get_vigil_count_total <- function(issue_name, ccc) {
  vigils <- ccc %>%
    filter(str_detect(issue_tags_summary, issue_name)) %>%
    summarise(vigil_count = n()) %>%
    mutate(issue = issue_name)

  return(vigils)
}

military_vigils_total <- get_vigil_count_total("military", ccc)
guns_vigils_total <- get_vigil_count_total("guns", ccc)
racism_vigils_total <- get_vigil_count_total("racism", ccc)
policing_vigils_total <- get_vigil_count_total("policing", ccc)
education_vigils_total <- get_vigil_count_total("education", ccc)

vigil_data_total <- bind_rows(military_vigils_total,
                              guns_vigils_total,
                              racism_vigils_total,
                              policing_vigils_total,
                              education_vigils_total)

vigil_data_total$issue <- factor(vigil_data_total$issue,
                                 levels = vigil_data_total$issue[order(vigil_data_total$vigil_count, de
```

```r
issue_colors <- viridis(5, option = "inferno")

vigil_data_total$color <- case_when(
  vigil_data_total$issue == "military" ~ issue_colors[1],
  vigil_data_total$issue == "guns" ~ issue_colors[2],
  vigil_data_total$issue == "racism" ~ issue_colors[3],
  vigil_data_total$issue == "policing" ~ issue_colors[4],
  vigil_data_total$issue == "education" ~ issue_colors[5]
)

ggplot(vigil_data_total, aes(x = issue, y = vigil_count, fill = issue)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Top Five Issues Protested at Vigils",
    x = " ",
    y = "Total Count",
    fill = " "
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  scale_fill_manual(values = issue_colors)
```




```r
ggsave("issuesbar.png", width = 10, height = 8, dpi = 300)
```

```r
get_vigil_monthly <- function(issue_name, ccc) {
  vigils <- ccc %>%
    filter(str_detect(issue_tags_summary, issue_name)) %>%
    mutate(date = as.Date(date, format = "%Y-%m-%d")) %>%
    mutate(year_month = floor_date(date, "month")) %>%
    group_by(year_month) %>%
    summarise(vigil_count = n(), .groups = "drop") %>%
    mutate(issue = issue_name)

  return(vigils)
}

military_vigils_monthly <- get_vigil_monthly("military", ccc)
guns_vigils_monthly <- get_vigil_monthly("guns", ccc)
racism_vigils_monthly <- get_vigil_monthly("racism", ccc)
policing_vigils_monthly <- get_vigil_monthly("policing", ccc)
education_vigils_monthly <- get_vigil_monthly("education", ccc)

vigil_data <- bind_rows(military_vigils_monthly,
                        guns_vigils_monthly,
                        racism_vigils_monthly,
                        policing_vigils_monthly,
                        education_vigils_monthly)

ggplot(vigil_data, aes(x = year_month)) +
  geom_line(aes(y = vigil_count, color = issue), size = 1) +
  geom_point(aes(y = vigil_count, color = issue), size = 2) +
  labs(title = "Top Five Issues of Vigils Over Time",
       x = " ",
       y = "Number of Vigils", subtitle = "By month from 2020 to 2025") +
  scale_color_manual(values = issue_colors) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  theme_minimal() +
  theme(legend.title = element_blank())
```




```r
ggsave("vigilsovertime.png", width = 10, height = 8, dpi = 300)
```